The H1 Occurrence-Crosswalk Route Decision: Canonical Shadows, Conservative Completion Obstructions, and Theorem-Backed Witnesses

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Abstract

The complete native cut leaves two nonsoft classes: eligible-tail-open and frontier-unmapped paths. In the frozen $X = 512$ archive there are 0 paths of the first type and 2,988 of the second, but the growing-scale contract quantifies over their disjoint union. TPC-153 constructs the canonical conservative shadow determined by the archived cut fields. TPC-154 proves that the same shadow admits inequivalent formal conservative completions, so it cannot select actual canonical parents, downstream stages, physical occurrences, or one-to-many multiplicities. TPC-155 supplies an exact verifier for a future theorem-backed occurrence witness, while correctly reporting that no production witness is present.

We convert these results into a typed alternative-route decision. The map clause requires a theorem-backed occurrence crosswalk, all nine four-map defects to vanish, and an independently complete physical-occurrence registry. The scalar clause requires both a complete original-scale $o(X)$ theorem for every frontier-unmapped path and a theorem-backed disposition of every eligible-tail-open path. Thus

$$\mathsf{H1Complete} = \mathsf{MapClause} \ \vee \ \mathsf{ScalarClause}.$$

For the source-locked TPC-151–155 snapshot, both clauses are NOT-TESTABLE. The current-schema-only construction is stopped in its declared cell, while the occurrence-augmented and scalar-plus-ETO routes remain open. The minimal missing node on the selected map route is refined to

$$\boxed{\mathsf{H1.theorem\_backed\_occurrence\_provenance\_crosswalk}}.$$

This is structural L1 progress and a sharper analytic target, not a positive L2, endpoint, prime-pair, or twin-prime result.

Keywords: fixed shift; occurrence provenance; conservative lift; typed alternative route; source-locked decision.

Claim boundary. The canonical cut-occurrence shadow is not the actual downstream occurrence lift. Formal completion nonuniqueness stops only a current-artifacts-only derivation; it is not an impossibility theorem on the unknown actual carrier. An executable witness verifier does not manufacture a theorem-backed production witness. No positive fixed-X-power L2 estimate, 1/400 endpoint, complete physical carrier, prime-pair lower bound, or twin-prime theorem is claimed.

1 The exact imported state

Let

$$\mathcal{C}_X^{\mathrm{ns}} = \mathcal{C}_X^{\mathsf{ETO}} \sqcup \mathcal{C}_X^{\mathsf{FUM}}$$

be the nonsoft part of the complete TPC-136 cut archive. The two terminal tags mean *eligible tail open* and *frontier unmapped*, respectively. Every growing-scale statement below quantifies over both classes. The finite census

$$|\mathcal{C}_X^{\mathsf{ETO}}| = 0, \qquad |\mathcal{C}_X^{\mathsf{FUM}}| = 2{,}988$$

does not prove that $\mathcal{C}_X^{\mathsf{ETO}}$ is asymptotically empty [2, 4].

TPC-153 copies every literally sourced cut field into a separate namespace

$$\mathcal{O}_X^{\mathrm{cut}} = \{\widehat{c} : c \in \mathcal{C}_X^{\mathrm{ns}}\}$$

and defines

$$L_X^{\mathrm{cut}} e_c = e_{\widehat{c}}.$$

It proves

$$\mathbf{1}^T_{\mathcal{O}_X^{\mathrm{cut}}} L_X^{\mathrm{cut}} = \mathbf{1}^T_{\mathcal{C}_X^{\mathrm{ns}}}.$$

The shadow preserves the cut identifier, native tuple, terminal tag, dyadic multiplier, h_0, physical normalization, weight source, boundary data, support role, and integrity lineage. It supplies no downstream parent or occurrence field [5].

If a candidate actual completion has source map $s : \mathcal{O}_X \to \mathcal{C}_X^{\mathrm{ns}}$, let

$$q : \mathcal{O}_X \longrightarrow \mathcal{O}_X^{\mathrm{cut}}, \qquad q(o) = \widehat{s(o)}$$

and let q_* be its zero-one pushforward matrix. The universal shadow identity is

$$q_* L_X = L_X^{\mathrm{cut}}. \tag{1.1}$$

It says that each cut column is returned exactly after all of its row-separated children are forgotten. It does not determine those children.

TPC-154 constructs inequivalent formal completions satisfying (1.1): a one-child completion with coefficient 1, a two-child completion with coefficients $1/2, 1/2$, and distinct legal parent/determinant labels. These examples live in the maximal formal class constrained only by archived equations. They prove current-schema nonidentifiability, not actual-support multiplicity [6].

TPC-155 then specifies a production witness contract. Its synthetic records exercise both terminal types, one-to-one and one-to-many edges, signed exact coefficients, separate cover and reconnection records, and a nonduplicated occurrence registry. They are L0 verifier regressions only. The production declaration is

$$\mathsf{production\_witness\_present} = \text{false}, \qquad \mathsf{actual\_witness} = \text{NOT-TESTABLE}$$

[7].

2 The theorem-backed crosswalk

Definition 2.1 (Occurrence-provenance crosswalk)**.** For every $c \in \mathcal{C}_X^{\mathrm{ns}}$, a theorem-backed occurrence-provenance crosswalk supplies a finite nonempty row set $\mathcal{R}_c$. A row $r \in \mathcal{R}_c$ contains:

(i) a unique row identifier, source cut identifier, target occurrence identifier, physical occurrence identifier, physical group identifier, and exact coefficient λ_r;

(ii) the native tuple, h_0, physical normalization, cut multiplier, boundary and mask data;

(iii) one canonical parent, its inverse-aggregation relation and computational and physical multiplicities;

(iv) determinant-fiber and ordered zero-mode records, including the literal map edges;

(v) an ordered stage path, each local multiplier, and source and target shift tags;

(vi) distinct formal-envelope, canonical-parent, and actual-active support statuses;

(vii) physical-cover and reconnection destinations; and

(viii) trusted source identifiers for every nontrivial provenance, equality, and nonzero assertion.

It is conservative when

$$\sum_{r\in\mathcal{R}_c} \lambda_r = 1 \qquad (c \in \mathcal{C}_X^{\mathrm{ns}}) \tag{2.1}$$

as an exact symbolic identity.

The rows, mathematical occurrences, and physical groups are different namespaces. Several rows may reach one mathematical or physical occurrence; they are not merged before (2.1) is checked. Cancellation between different source columns is forbidden. Exact rational, signed, or Gaussian-rational coefficients are allowed; floating-point equality is not.

The native coordinate called d is not the affine intercept also commonly called d. A crosswalk that uses both must contain an explicit theorem-backed relation between them. Likewise, copying the cut value h_0 into a later record is not a proof that an unarchived stage preserves the prescribed shift.

The stage names in the TPC-141 physical-cost registry are administrative ledger occurrences. They do not attach a cut row to a downstream parent, target, or local multiplier, and therefore cannot be reused as row-level stage edges [1].

Definition 2.2 (Trusted production witness)**.** A crosswalk is a trusted production witness when:

(a) its source domain is exactly $\mathcal{C}_X^{\mathsf{ETO}} \sqcup \mathcal{C}_X^{\mathsf{FUM}}$;

(b) every row has one source and a complete ordered provenance chain;

(c) (2.1), native metadata fidelity, and normalization fidelity hold;

(d) every source identifier belongs to a source-locked trusted theorem registry; and

(e) cover, reconnection, and occurrence-registry completeness are supplied and validated as three distinct exports.

Remark 2.3 (Verifier soundness boundary)*.* An executable verifier checks representation, exact algebraic identities, completeness, and source locks. It cannot establish the external mathematical truth of a self-asserted source label. Accordingly, a source identifier outside the trusted registry is a failure rather than evidence.

3 The typed H1 alternative

The earlier phrase "complete the four maps" hides cover, reconnection, and registry obligations. We keep every component explicit. Let

$$\mathfrak{D}_X = (D_L, D_{Q_D}, D_{Q_Z}, D_G, D_P, D_{DZ}, D_{GP}, D_{\mathrm{cover}}, D_{\mathrm{rec}})$$

be the TPC-146 defect vector. Let D_{occ} be the independent occurrence-registry completeness defect.

Definition 3.1 (Map clause)**.** The map clause is

$$\mathsf{MapClause}(X) := \mathsf{TrustedCrosswalk}(X) \;\wedge\; \bigwedge_{D\in\mathfrak{D}_X} (D=0) \;\wedge\; (D_{\mathrm{occ}}=0).$$

Thus a valid crosswalk only makes the descendants evaluable. It does not automatically prove that all their defects vanish. In particular, an aggregate commuting square cannot replace the row-separated fixed-h_0 test.

Definition 3.2 (Scalar-plus-ETO clause)**.** The scalar clause is

$$\mathsf{ScalarClause}(X) := \mathsf{CompleteFUMSoft}(X) \;\wedge\; \mathsf{TheoremBackedETODisposition}(X).$$

The first factor means a complete original-scale $S_{\mathrm{front}}(X) = o(X)$ theorem on every FUM path. The second means totalization or a complete soft theorem on every ETO path at growing scales.

An empty ETO class in one fixture cannot prove the second factor. Likewise a theorem on one convenient frontier subfamily cannot prove the first.

Theorem 3.3 (Typed H1 completion criterion)**.** *Within the declared TPC-146 frontier architecture,*

$$\boxed{\mathsf{H1Complete}(X) \iff \mathsf{MapClause}(X) \vee \mathsf{ScalarClause}(X).}$$

The clauses are alternatives, not two halves of one conjunction. This is the explicit typed form of the two-route contract introduced in TPC-146 [3].

Proof. The map clause is precisely the row-resolved completion route: it requires a valid lift, literal determinant and zero-mode maps, physical grouping, prescribed-shift localization, both compatibility tests, cover, reconnection, and a complete occurrence registry. The scalar clause removes the complete FUM contribution while separately disposing of ETO. These are the two sufficient routes declared in TPC-146. Neither route can borrow a missing factor from the other. □

4 Current route decision

Proposition 4.1 (Current statuses)**.** *For the source-locked TPC-151–155 production snapshot:*

$$\begin{aligned}
\mathsf{H1.cut_occurrence_shadow} &= \textsc{proved}_{\mathrm{L1}\text{-}structural}, \\
\mathsf{H1.current_artifacts_only_canonical_actual_lift} &= \textsc{stop-declared-route}, \\
\mathsf{H1.theorem_backed_occurrence_provenance_crosswalk} &= \textsc{not-testable}, \\
\mathsf{MapClause} &= \textsc{not-testable}, \\
\mathsf{ScalarClause} &= \textsc{not-testable}, \\
\mathsf{H1Complete} &= \textsc{not-testable}.
\end{aligned}$$

The occurrence-augmented map route and the scalar-plus-ETO route remain open.

Proof. TPC-153 supplies the first status. TPC-154 proves the scoped current-artifacts-only obstruction. TPC-155 records no production actual witness, so D_L and every downstream defect remain unevaluable. Neither a complete FUM scalar theorem nor a growing-scale ETO disposition is imported. The two conjunctions in definitions 3.1 and 3.2 therefore fail testability, and theorem 3.3 gives the final status. □

Theorem 4.2 (Refined minimal missing node)**.** *On the selected occurrence-augmented map route, the minimal missing antichain is the singleton*

$$\boxed{\{\mathsf{H1.theorem\_backed\_occurrence\_provenance\_crosswalk}\}.}$$

The actual occurrence lift, all four literal maps, cover, reconnection, and registry totality are descendants.

Proof. The canonical shadow is already proved. The next edge of the typed DAG requires a production crosswalk satisfying definition 2.1; no source-locked artifact supplies one. Each map-route descendant needs rows or fields created by that crosswalk. Hence none is a smaller missing ancestor. □

Remark 4.3 (Two antichains, not one false conjunction)*.* The scalar route has its own minimal missing antichain,

$$\{\mathsf{H1.complete\_FUM\_scalar\_oX}, \mathsf{H1.theorem\_backed\_ETO\_disposition}\}.$$

The executable snapshot stores both clause-wise antichains and chooses a canonical representative only inside the selected clause.

5 Scoped stops and kill criteria

Route cell	Current state	Exact criterion
Current-artifacts-only canonical derivation	STOPPED	Stopped by conservative formal-completion nonuniqueness.
Occurrence-augmented map	NOT-TESTABLE, open	Passes only with a trusted production crosswalk, zero map defects, complete cover and reconnection, and a complete registry.
Scalar plus ETO	NOT-TESTABLE, open	Passes only with complete original-scale FUM $o(X)$ and a theorem-backed ETO disposition.
Whole architecture	not stopped	Infeasibility additionally requires an independent theorem that the route universe is complete.

The map route fails closed if a nonsoft source is deleted or duplicated, if ETO is omitted because the current fixture is empty, if conservation is checked only after cross-source aggregation, if parent or stage data lack a trusted source, if native, h_0, or normalization metadata drift, or if an unevaluated coefficient is promoted to actual active support.

The scalar route fails closed if either conjunct in definition 3.2 is missing. A finite ETO census is not a disposition theorem. A declared list of failed routes is not a complete route universe.

6 Executable source-locked audit

The standard-library program `experiments/tpc156_h1_occurrence_decision.py` runs the TPC-152–155 source audits in read-only mode, imports only allowlisted logical artifacts, canonicalizes text as `CANONICAL_UTF8_LF_V2`, and records SHA-256 locks. Hashes have integrity semantics only; they are not mathematical proofs.

The generated decision manifest stores:

(i) exact production counts and imported statuses;

(ii) a typed `ALL/ANY_CLAUSE` DAG;

(iii) the nine four-map defects and the separate registry defect;

(iv) clause-wise minimal missing antichains;

(v) scoped route stops and open alternatives; and

(vi) strict no-promotion claims.

Mutation tests reject an identity-shadow promotion, omission of the ETO domain, a compressed four-map claim that hides cover, reconnection, or registry totality, a scalar-only route pass, an ETO-only route pass, finite-empty-ETO reasoning, an unsupported actual-active label, and an architecture-infeasible verdict without a route-universe theorem. Positive hypothetical scenarios verify that either a complete map clause or both scalar-clause factors would close H1. They are classifier regressions, not production evidence.

7 Decision theorem and next analytic object

Theorem 7.1 (TPC-156 route decision)**.** *For the canonical source-locked production artifacts through TPC-155,*

$$\boxed{\mathrm{Verdict}_{H1} = \text{NOT-TESTABLE.}}$$

The current-schema-only actual-lift cell is stopped. The selected occurrence-augmented map route remains open, with first missing object the theorem-backed occurrence-provenance crosswalk of definition 2.1. The scalar-plus-ETO alternative also remains open with both of its factors missing.

Corollary 7.2 (Exact progress level)**.** *TPC-153–156 establish a canonical conservative shadow, a scoped formal-completion obstruction, a production witness contract, and a typed route decision. This is genuine structural* L1 *progress. It does not establish a complete actual physical carrier or any positive arithmetic* L2 *estimate.*

The next positive map-route paper must derive, rather than merely schema-complete, the row-level crosswalk. In particular, it must explain how every ETO and FUM cut path reaches its canonical parent and ordered downstream stages, why the exact child coefficients sum to one, and which theorem sources justify active support, cover, and reconnection. If this derivation fails, the exact scoped negative result remains reusable and the scalar-plus-ETO route is still available.

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